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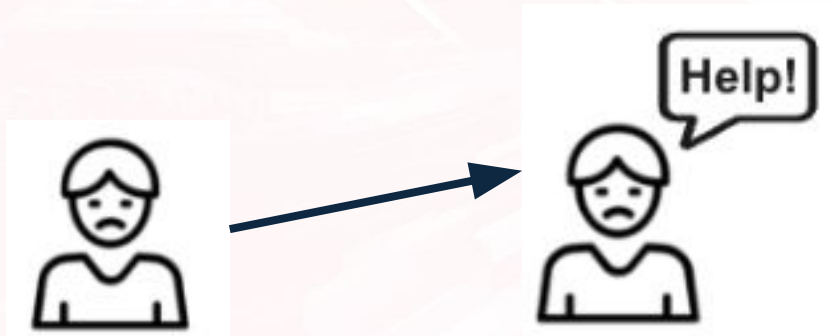
WIP: Automated Flexible Extensions for Improving Learning Equity in Large Scale Computing Classrooms

Dana Benedicto*, Jordan Schwartz*, Narges Norouzi, Lisa Yan

Case Study: UC Berkeley's CS and DS courses



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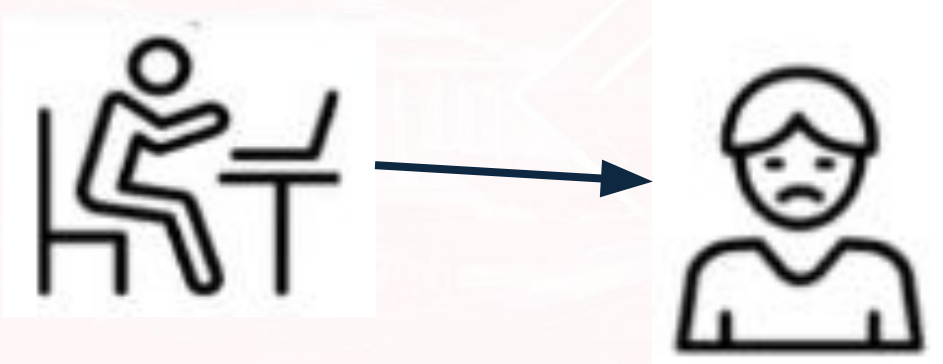
Case Study: UC Berkeley's CS and DS courses



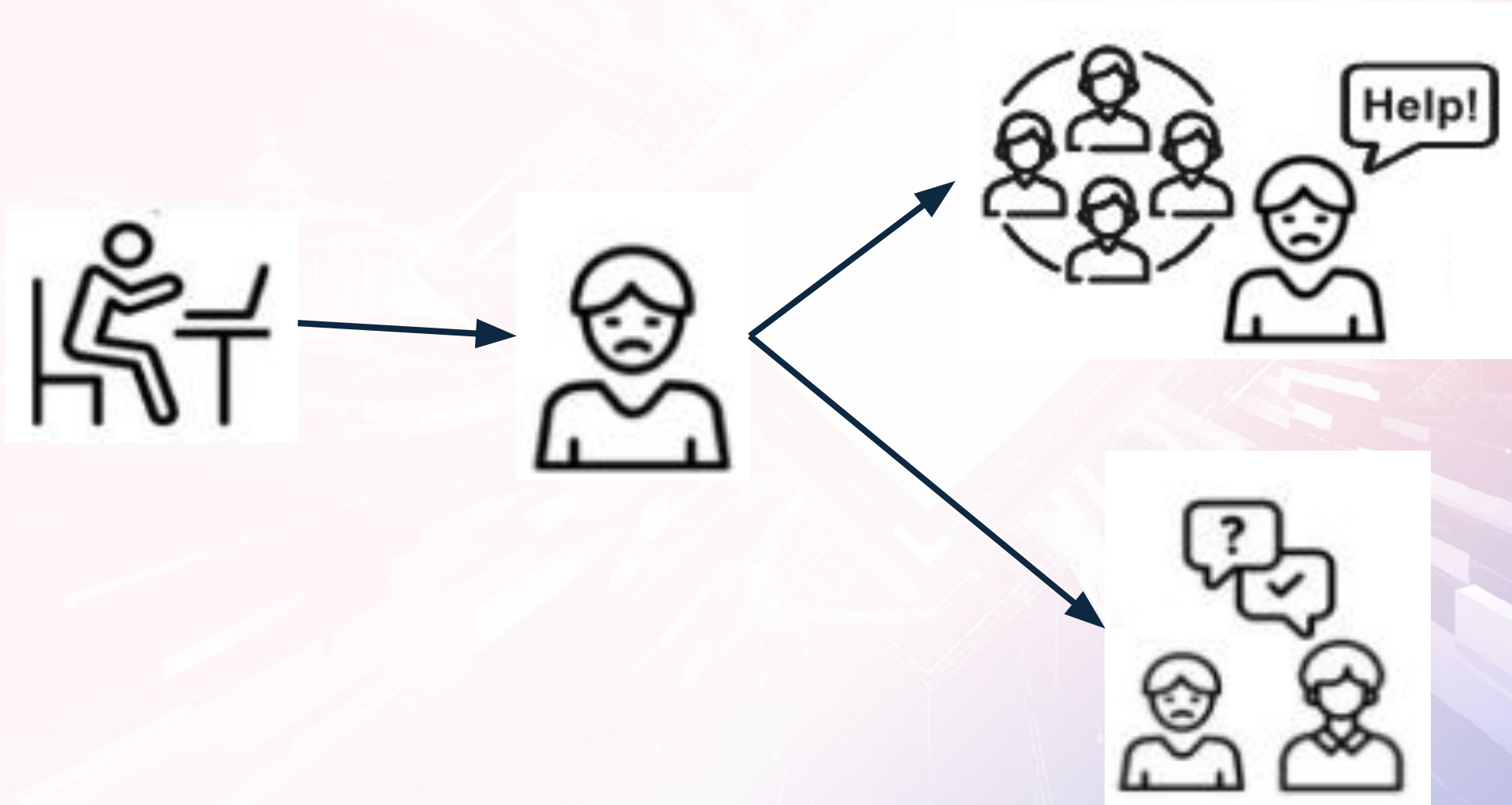
Flexible Classrooms - “Flexextensions”



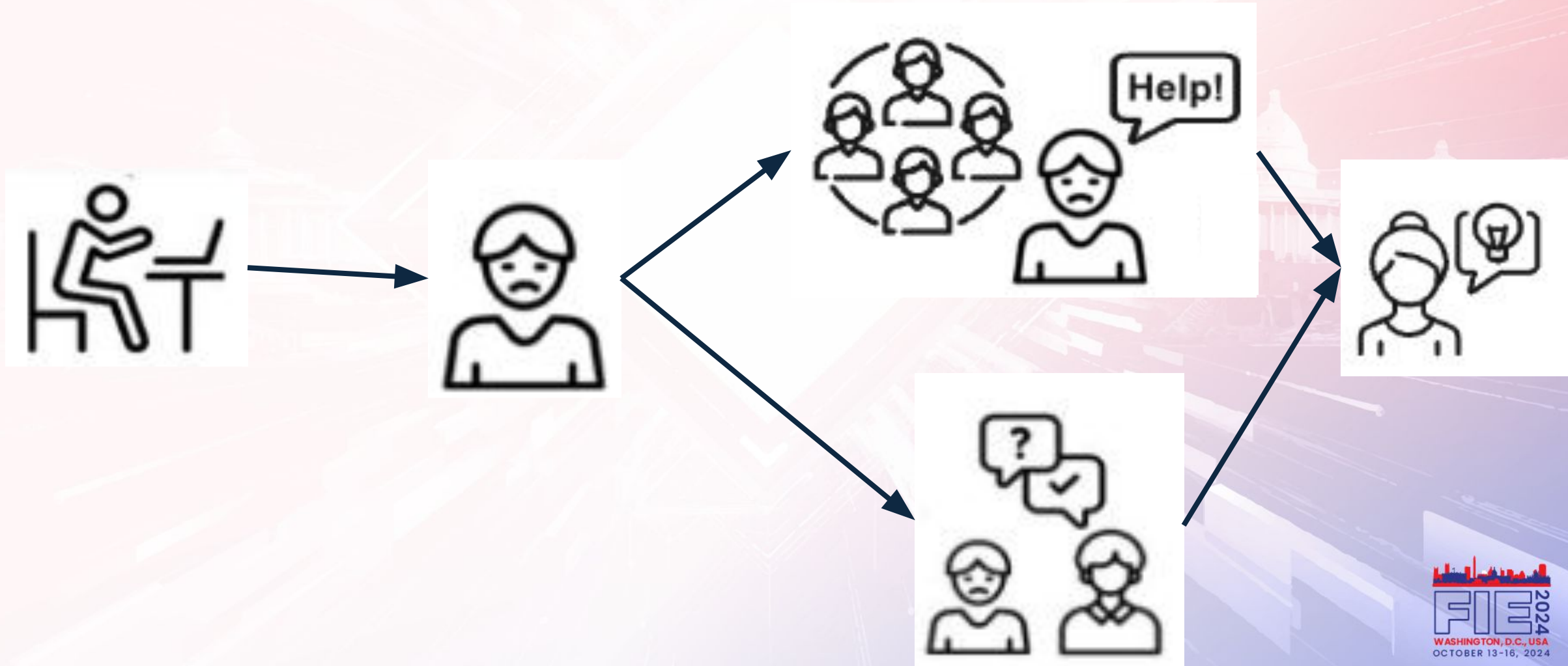
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The flexextensions tool reduces the barrier to asking for help with minimal instructor overhead

Students feel valued as individuals

“The extension policy in this class is the only one that made me feel like this class understood what it was like to be a student.”

“Extensions saved my life and mental health.”

“[... flexextensions] severely reduced stress around deadlines and made me feel confident that the course cared about my well being as a person not just a student.”

“Allowing extensions helps students to learn better instead of just googling answers to meet a deadline.”

Agenda

Google based tool walkthrough

- Google Form student perspective
- Google Sheet staff perspective

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3 course based case study

- Across CS and DS
- Why do students request extensions?

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Limitations and Future work

- What did we do well?
- What could we improve on?

Student Perspective: Students: Fill out a Google Form



Student support is integrated into course structure; students request extensions via a Google form



Google Forms

Question	Description
What is your student ID?	Used to create the extension. Emails are automatically collected.
What is your name?	Used to track students.
(OPTIONAL) Are you registered with the disabled students program?	Some courses choose to include this depending on whether their extension policy takes into account whether a student is registered with programs and services for disabled students.
Which assignment would you like an extension on?	All assignments eligible for an extension are listed here for students to choose from.
How many days would you like an extension for?	Students must submit the total number of days they would like an extension for. The system uses absolute offsets. If a student selects multiple assignments in the previous question, they can comma-separate the number of days for each assignment request, a regular expression verifies the response. Students can also request a longer extension for an assignment they've already requested an extension on by putting in a larger offset.
Why do you need this extension?	How courses enforce this question varies based on their policy. This is a free response text question.
Are you working with a partner?	The automation supports emailing and extending for both partners on one submit request.
What is your partner's email?	Used to email the partner.
What is your partner's student ID?	Used to create the extension.

Course Staff Perspective: See all requests at once

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	email	is_dsp	hw1	hw2	p1-cp	p1	p2	p3	email_commen	approval_status	email_status	notes	flush_gradesco	last_run_outpu	last_run_reaso	last_run_times
2	emailA@institution.edu		10							Manually App	In Queue		☐	Auto-approved.	sick	8/23/2023 3:38:2
3	emailB@institution.edu		2							Auto Approve	Auto Sent		☐	Auto-approved.	midterms	9/29/2023 2:45:4
4	emailC@institution.edu	Yes		5	5					Auto Approve	Auto Sent		☐	Auto-approved.	stressed	10/4/2023 10:01
5	emailD@institution.edu				3				This extension re	Auto Approve	Auto Sent		☐	Auto-approved.	covid	10/7/2023 15:40
6	emailE@institution.edu			6						Auto Approve	Auto Sent	met with on 9/28	☐	Auto-approved.	family emergenc	9/29/2023 22:04
7	emailF@institution.edu		2	3						Manually App	Auto Sent		☐	A dsp request of	tired	10/9/2023 0:19:2
8	emailG@institution.edu	Yes		10						Pending	Pending App		☐	A dsp request of	DSP	9/28/2023 19:14
9	emailH@institution.edu		20							Manually App	Auto Sent		☐	Student requeste	i forgot to submit	9/2/2023 20:34:5

extension-bot



extensionbot APP Yesterday at 1:00 PM
 @Jordan Schwartz An extension request needs review (student requested a retroactive extension on an assignment).

Email: jordanschwartz@berkeley.edu>jordanschwartz@berkeley.edu

Assignment(s): Homework 1

Reason: Need more time

DSP Accomodations for Extensions: Yes

Partner Email(s): yuerou.tang@berkeley.edu

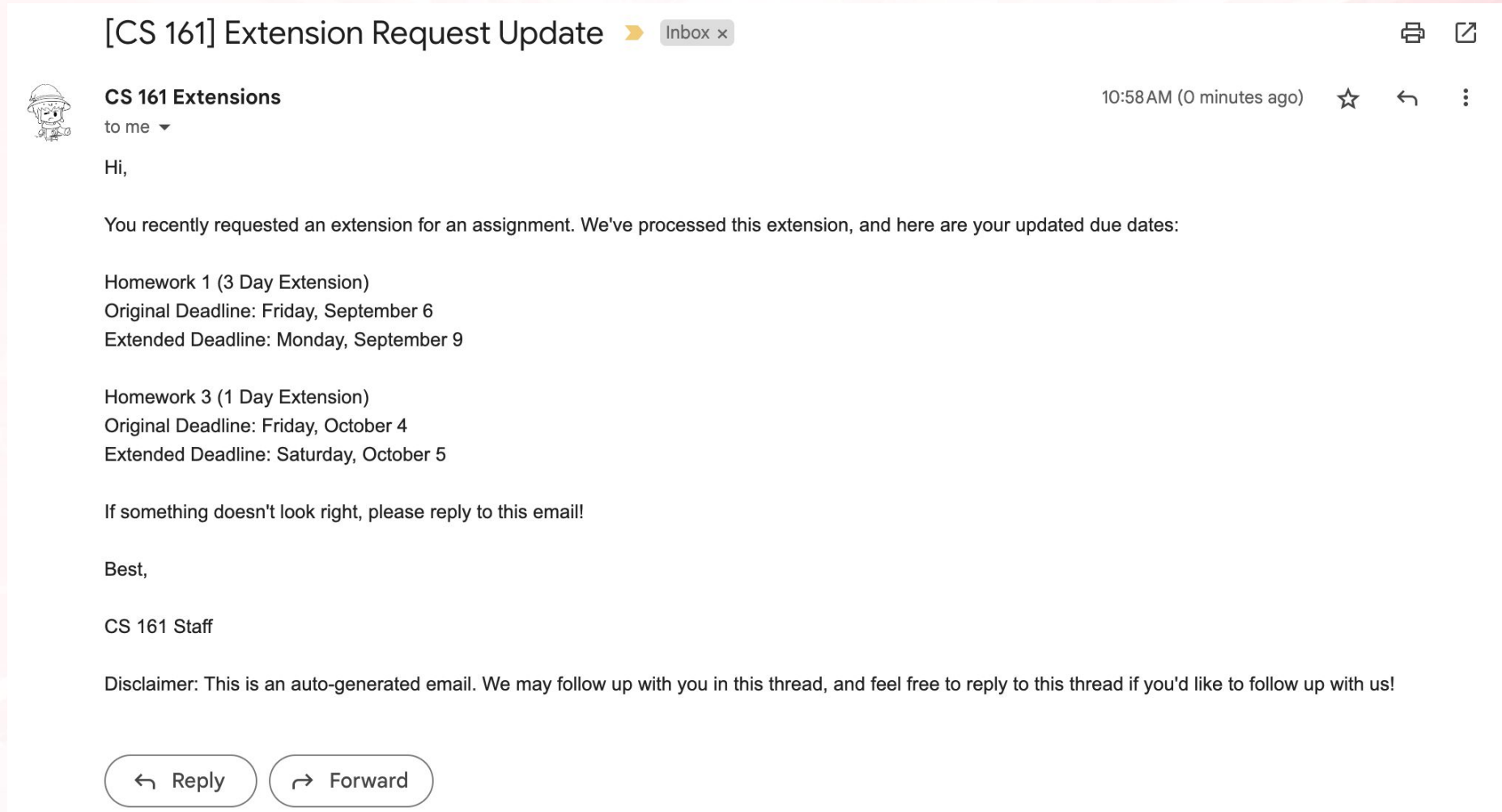
Homework 1 1

Warnings:

[Homework 1] student requested an extension for 1 days, which was <= an existing request of 1 days, so we kept the existing request in-place.

[View Spreadsheet](#)

Course Staff Perspective: Automated Emails



Case Study: 3 courses

Data 1

- freshman level
- accepts some late work with penalties
- drop 2 lowest HWs and labs

Data 2

- sophomore/junior level
- 1 day grace period

Data Engineering

- senior level
- accepts some late work with penalties
- 9 slip days across all assignments

All 3:

- automatically grant 2-day extensions to students registered with the campus accessibility program, 0 for others

How Do Students Use Extensions?

2,633 students × (25 Data 1 Assignments
+ 27 Data 2 assignments
+ 9 Data Engineering assignments)
= 160,613 opportunities to request an extension

1,914 total extension requests / 160,613 opportunities
= 1.19% of all opportunities were used

The majority of students used the policy on only one assignment

Students were not requesting extensions frivolously just because they were given the opportunity to do so.

Survey Methodology

Student Extensions Requests

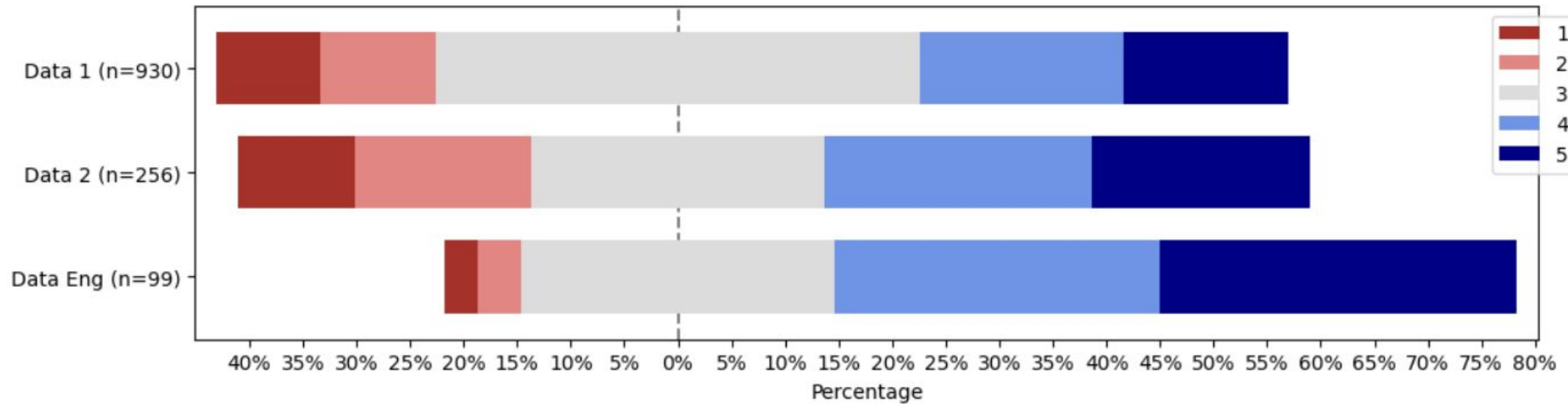
Extensions Roster

Course Evaluations

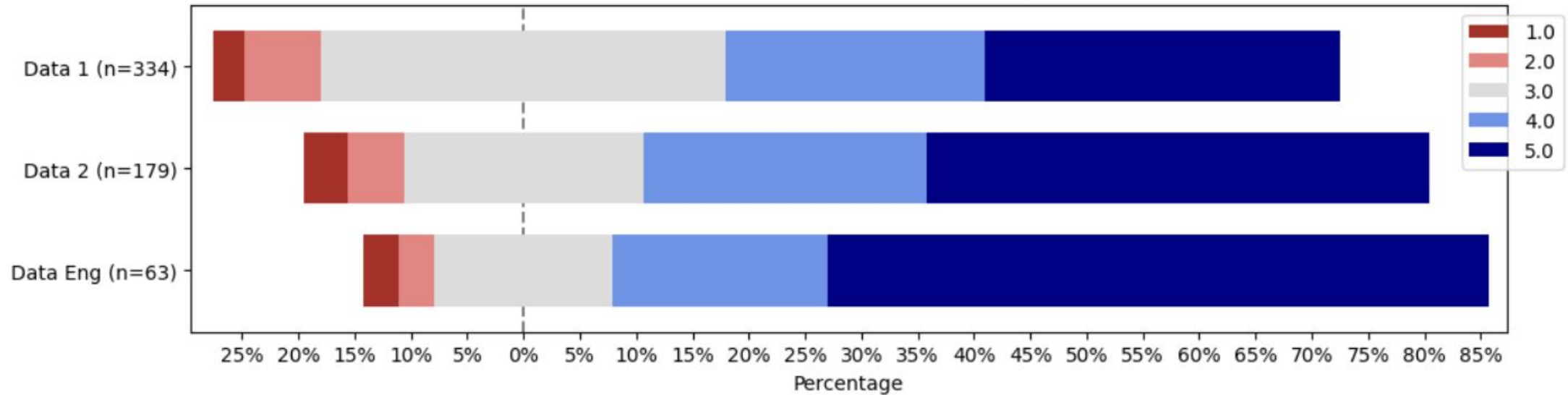
52% of all students in all 3 classes responded

How comfortable do you feel making an extension request?

(1.0 - Least Comfortable, 5.0 - Most Comfortable)

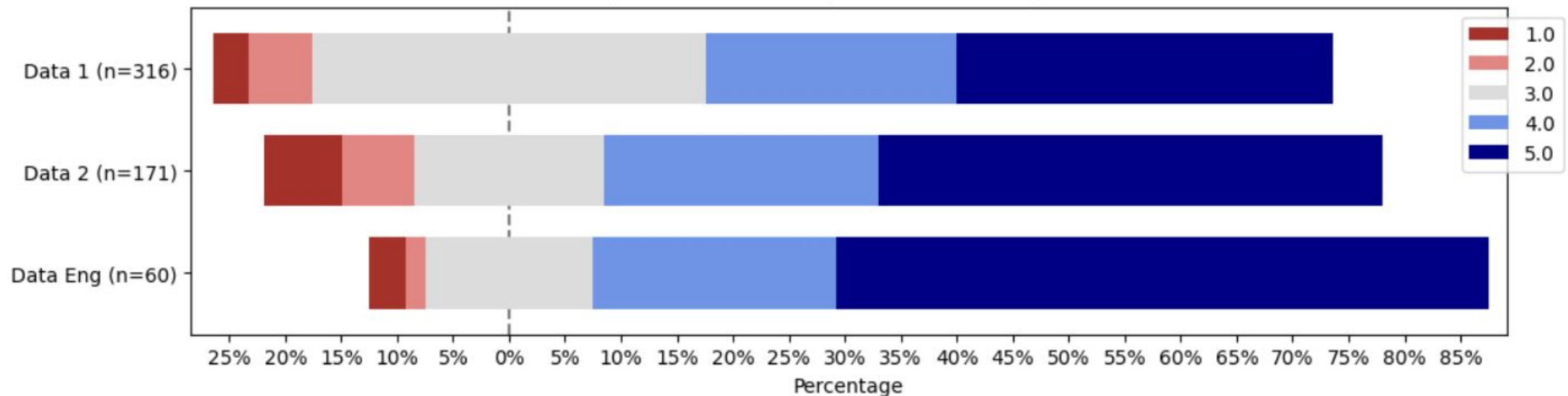


Do you think our extensions policy was valuable to your learning? (1.0 - Least Valuable, 5.0 - Most Valuable)



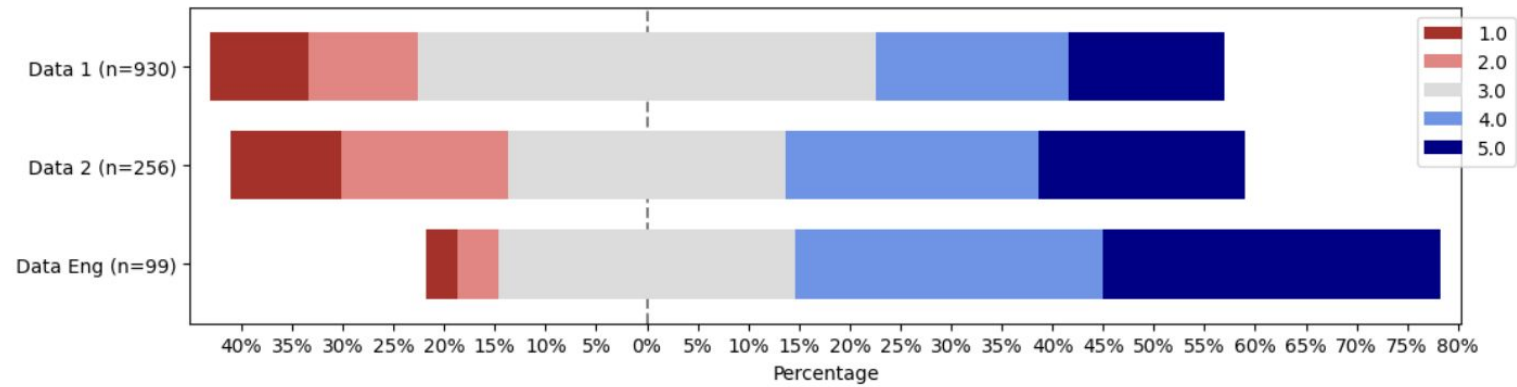
Do you think our extensions policy benefited your overall experience taking this class, or managing your overall academic experience this semester?

(1.0 - Least Beneficial, 5.0 - Most Beneficial)



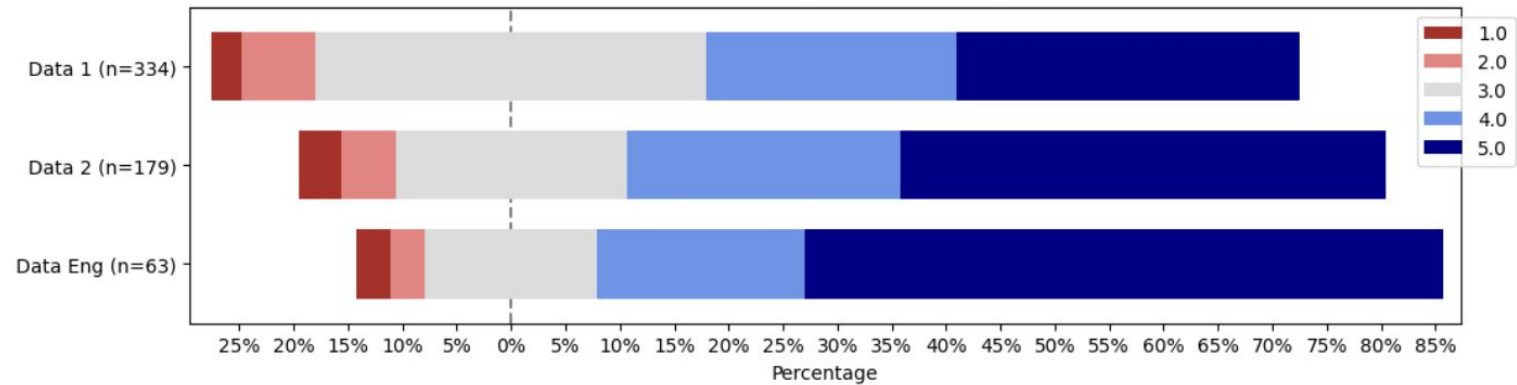
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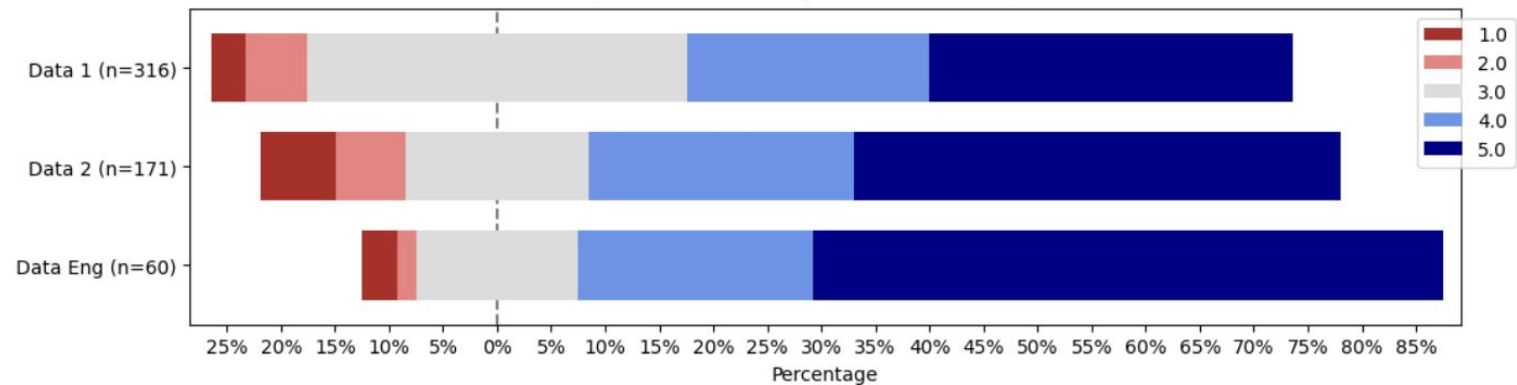
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Why Do Students Request Extensions?

TABLE I
COUNT AND PROPORTION OF REQUESTS UNDER EACH REASON.

Theme	Count	%
Medical	521	38.00
Family/Personal Circumstance	268	19.55
Workload	192	14.00
Catch-up Time	143	10.43
Mental Health	77	5.62
Technical Issues	86	6.27
Disability	34	2.48
Extracurricular	13	0.95
Other	37	2.70

Why Do Students Not Request Extensions?

TABLE II
COUNT AND PROPORTION OF NON-REQUESTS UNDER EACH REASON

Theme	Count	%
Feelings of doubt, shame, or being undeserving	36	25.71
Fear about being rejected, perceived as lazy or not hardworking	29	20.71
Negative sentiment due to past experience/syllabus wording	29	20.71
Uncertainty or lack of awareness	26	18.57
Personal desire to stick to assignment deadlines	20	14.29

How could we have improved our study?

- Why do we ask and require that students fill out a reason?
 - leads to fabricated reasons
- Could we have improved this study?
 - Students self-selected to partake in the perception part of the study
 - All courses are CS or DS, but not all students are necessarily those majors
- Should TAs do these support meetings or instructors?
 - Often done by TAs at scale
 - trust and consistency

Plans to Expand and Improve

- Expand to further classrooms,
 - outside of CS and DS
 - outside of Berkeley
- Long term academic impacts of *Flexextensions*